# RGC — Full Stack Intern Technical Assessment

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| **Role** | Full Stack Intern |
| **Time limit** | 24 hours from receiving the task |
| **Data files** | transcripts.json, users.json, products.csv, brands.csv |
| **Deliverables** | Hosted app, Repository/source code, Submission-Notes, Working log |
| **Focus** | Full-stack delivery, data enrichment, retail/category insight, commercial storytelling |

## Context

At RGC, we work with product, retail, SKU, reviewer, packaging, brand, and AI-enriched data to help brands understand markets, products, and customers.

For this assessment, we want to see how you approach a realistic, data-centred product problem. We are not looking for a perfect production system. We are looking for clear thinking, thoughtful prioritisation, good technical foundations, and the ability to ship something useful within a short timebox.

## The Task

Build a hosted full-stack dashboard application that helps a brand, product, retail, or insights team understand the provided category dataset.

The dataset combines two broad types of signals:

1. Consumer voice signals from reviewer transcripts, ratings, user profiles, and evidence-backed suggested tags.
2. Retail, product, and brand intelligence signals from the enriched products and brands data.

Your job is not simply to display the files separately. Your job is to turn the data into a useful intelligence tool: something that helps a commercial user understand what consumers are saying, how products and brands are positioned, how brands compare within the category, and what opportunities or tensions may be worth investigating.

In your submission notes, please include a brief section explaining what you would do with more time. This can include product or technical improvements, as well as any additional data you would want access to, what that data might contain, and how you would use it to improve the dashboard or generate stronger insights.

***Note****: You do not need to use every column in the provided data. The dataset contains many columns because we want to see how you prioritise different types of information, choose which signals matter and use the data creatively.*

## Who is this dashboard for?

Imagine this as a client-facing dashboard inside the RGC platform, built from an anonymised product/reviewer/category dataset.

The user is a commercial, product, insights, retail, or category team at a brand. They want to understand consumer perception, product strengths and weaknesses, usage occasions, purchase drivers, competitor context, and opportunities worth investigating.

For this assessment, **do not spend time** on authentication, permissions, or client-specific data access. Treat the provided dataset as a shared anonymised sample designed to show what a useful brand-facing insight experience could look like.

## Provided Dataset

You will receive four data files. The files are a mix of JSON and CSV. You are not expected to use every field. We care more about thoughtful selection, joining, and explanation of signals than exhaustive display of every column.

| File | Records | Purpose |
|---|---|---|
| **transcripts.json** | 130 | Reviewer transcript records, including review text and related review-level information. |
| **users.json** | 130 | Anonymised reviewer profile information, including demographics, behavioural tags, shopper archetypes, and onboarding answers. |
| **products.csv** | 35 | Product-level retail and AI-enriched information, including product metadata, pricing, ingredients, claims, positioning, usage scenarios, target users, product attributes, labels, and category context. |
| **brands.csv** | 9 | Brand-level intelligence, including brand context, performance signals, social/web metrics, audience/archetype information, and wider category benchmark data. |

The files can be joined using shared identifiers and fields. Transcripts link to products using productId. Users link to transcripts using reviewId and personId. Products link to brands using the brand field.

### Important note on transcript coverage

Transcript coverage is intentionally partial. Some brands and products have reviewer transcripts, while others are included without transcripts to provide wider retail, category, and competitive context.

In this dataset, 15 products have linked reviewer transcripts. The remaining catalogue-only products still include product, retail, enrichment, and brand intelligence fields. Missing transcript data should not be treated as an error.

Use transcript-backed brands to understand consumer voice. Use the wider product and brand data to understand category context, competitor positioning, and retail comparisons. A strong dashboard may connect these two views, rather than treating them as separate datasets.

### About products.csv and brands.csv

products.csv is not just basic product metadata. It includes standard retail fields such as name, brand, category, price, pack size, ingredients, retailers, and imagery, as well as enriched fields such as health and dietary attributes, environmental and packaging attributes, market positioning, target users, usage scenarios, companion products, and machine-generated labels with confidence scores.

brands.csv adds a brand-level intelligence layer, including brand context, performance and traction signals, social/web metrics, archetype affinity, and category benchmark information.

Some enriched fields are AI-generated or pipeline-derived. Treat these as useful directional signals rather than perfect ground truth. Strong submissions may compare these signals against transcript evidence where transcript data exists.

*To reiterate, please do not spend time trying to incorporate every column. There is no single correct answer. We are interested in what you choose to focus on, how you connect the available signals, and what useful insight you can create from the data.*

## Core Challenges

### 1. Extract useful transcript-backed signals

Some users already have broad profile tags. The transcript text may reveal additional useful signals that are not currently captured.

These do not need to be personality tags. They could be behaviours, usage occasions, routines, shopper needs, product preferences, retailer mentions, complaints, competitor comparisons, purchase drivers, co-consumption habits, or anything else you believe would help a brand understand the customer.

If your app suggests a tag, attribute, theme, or signal, it should show the evidence behind it. For example, the phrase or sentence in the transcript that supports the suggestion.

### 2. Build a commercial dashboard from joined signals

The dashboard should help someone understand what the data means and why it might matter commercially.

We are especially interested in how you combine consumer voice data with product, retail, and brand intelligence. For example, you might compare what products claim or are positioned for against what reviewers actually say, or use catalogue-only competitor products to provide pricing, category, attribute, or brand context.

A strong submission does not need to use every field. It should make deliberate choices about what to connect, what to ignore, and what commercial story the data supports.

## What You Should Build

- A hosted web application using the provided dataset.
- A dashboard view that surfaces useful summary insights.
- A way to browse or explore products, brands, users, transcripts, existing tags, and suggested signals.
- A data enrichment feature that suggests useful tags, attributes, themes, or behavioural signals from transcript text.
- Evidence for suggested transcript signals, such as the transcript sentence or phrase that supports them.
- Filtering, searching, or sorting that makes the data easier to explore.
- At least one brand-facing or commercially useful insight feature that joins transcript/user signals with product, retail, or brand intelligence.
- A README.md explaining how to run the project, your approach, assumptions, tradeoffs, AI/tool usage, and what you would improve with more time.

## Possible Directions

You are free to decide what is most valuable. Strong directions might include:

- Suggested consumer signals with confidence levels and transcript evidence.
- Product or brand views showing common themes, praise, complaints, usage occasions, user segments, or competitor context.
- Insight cards such as “what customers love”, “what frustrates users”, “where positioning matches consumer voice”, or “what to investigate next”.
- Comparisons by brand, product, retailer, rating, price tier, product attributes, category benchmark, or user segment.

## Required Deliverables

- Live hosted application URL.
- GitHub repository URL or zipped source code.
- README.md with setup instructions and notes on your approach.
- Submission notes as a pdf.
- A working log as a separate working-log file.

Submission notes should briefly explain what you built and how to use it. They should cover the main views, features, charts, tables, or insight cards in the app; the key insights or relationships you found in the data; how you connected transcript/user signals with product, retail, or brand intelligence; any assumptions, limitations, or tradeoffs; whether you used AI tools and how you verified the output; and what you would improve with more time.

## Working Log / Thought Process

We are interested not only in what you ship, but in how you think while building. Please keep a lightweight working log as you go.

We recommend using Wispr Flow to quickly dictate thoughts while you work, or any equivalent note-taking approach that lets you capture decisions, questions, blockers, tradeoffs, and changes in direction without slowing yourself down.

Your log does not need to be polished. We are looking for a genuine trail of your reasoning: what you prioritised, where you got stuck, what you tried, how you used AI/tools, what you changed your mind about, and what you would do next with more time.

## Time Limit

You will have 24 hours from receiving the assessment to submit your work. We do not expect you to spend the full 24 hours actively coding. Focus on a clear, useful application rather than trying to build every possible feature.

## Technology

Use whichever stack lets you ship a high-quality result quickly. React, Next.js, Node, Python, FastAPI, Flask, SQLite, Supabase, or any other sensible stack are all fine. The important thing is that the app runs, the data is handled correctly, and the approach is understandable.

## What We Care About

- A working, hosted application that is easy to use.
- Thoughtful handling of structured, nested, and unstructured data.
- Explainable enrichment: if a signal is suggested, the user should be able to see why.
- Smart joining of transcript/user signals with product, retail, brand, and category intelligence.
- Commercial thinking: what would a brand, category owner, retail team, or competitive brand actually want to learn?
- Insight storytelling: the application should help someone understand what action or decision the data might support.
- Clear technical decisions and sensible tradeoffs for the timebox.
- Readable code, a clear README, and a useful working log.

## What Not to Spend Too Much Time On

- Authentication, permissions, client-specific access rules, or user accounts.
- Pixel-perfect design.
- Complex infrastructure.
- Using every possible column or building every possible chart.
- Over-engineering the enrichment system if a simpler explainable approach works well.

## Submission Format

1. Live URL
2. Repository URL or zipped source code
3. Submission Notes
4. Working log / thought-process notes